

SatQuery AI — Geospatial Analysis Report

Remote Sensing Intelligence & Multimodal Evidence Verification

Job ID:	1795e108-efd5-48d3-a3d4-b99d065fff2a
Project ID:	e64343dc-4fb3-41c3-ac45-61635ae3beaf
Task Type:	OPTICAL_SAR_HEAD
Status:	RUNNING
Generated:	2026-09-26 11:33:25 UTC

1. Request Specification

Query / Prompt: "Classify land cover classes across multimodal Sentinel-1 SAR and Sentinel-2 optical imagery."

2. Factual Findings & Observations

Finding 1 (optical_sar_head_finding): Multimodal Optical-SAR classification verified with CROMA: Primary land-cover class is 'Inland wetlands' (97.4% confidence). Active classes: Inland wetlands (97.4%).

Measurements: protocol_version: 1.0, success: True, job_id: a1a7262f-fbcb-4065-9009-66b362f1808b, step_key: optical_sar_classification, attempt_id: 1, output_assets: {}, output_metadata: {'model': 'Optical-SAR Classification Head (CROMA ViT-B + MLP)', 'task': 'optical_sar_classification', 'primary_class': 'Inland wetlands', 'confidence': 0.9744, 'detected_classes': [{'class': 'Inland wetlands', 'confidence': 0.9744}], 'summary': "Multimodal Optical-SAR classification verified with CROMA: Primary land-cover class is 'Inland wetlands' (97.4% confidence). Active classes: Inland wetlands (97.4%).", 'text': "Multimodal Optical-SAR classification verified with CROMA: Primary land-cover class is 'Inland wetlands' (97.4% confidence). Active classes: Inland wetlands (97.4%).", 'modalities': ['Sentinel-1 SAR (2-band)', 'Sentinel-2 Optical (12-band)'], 'vram_allocated_mb': 777.58, 'note': 'Downstream land cover classification head trained on BigEarthNet-MM / SEN1-2 schema.'}, metrics: {'duration_ms': 2171.480894088745, 'peak_vram_mb': 777.5830078125, 'peak_ram_mb': None, 'device_name': 'cuda', 'cudnn_enabled': False}, error_code: None, error_message: None

3. Operational Boundaries & Limitations

- Observations are generated based on local hardware inference (RTX 5060 Laptop GPU).
- ChangeFormer outputs represent 2D building change masks trained on LEVIR-CD.
- UPerNet semantic segmentation reflects generic ADE20K demo proxy classes unless domain-trained.
- Satellite pixel coordinates are not mapped to geographic bounds unless explicit CRS metadata is verified.
- Analyst confirmation is recommended prior to operational deployment.